

Podscape: Exploring the Comfort Level with Pods in Pedestrian Spaces through Immersive Simulation

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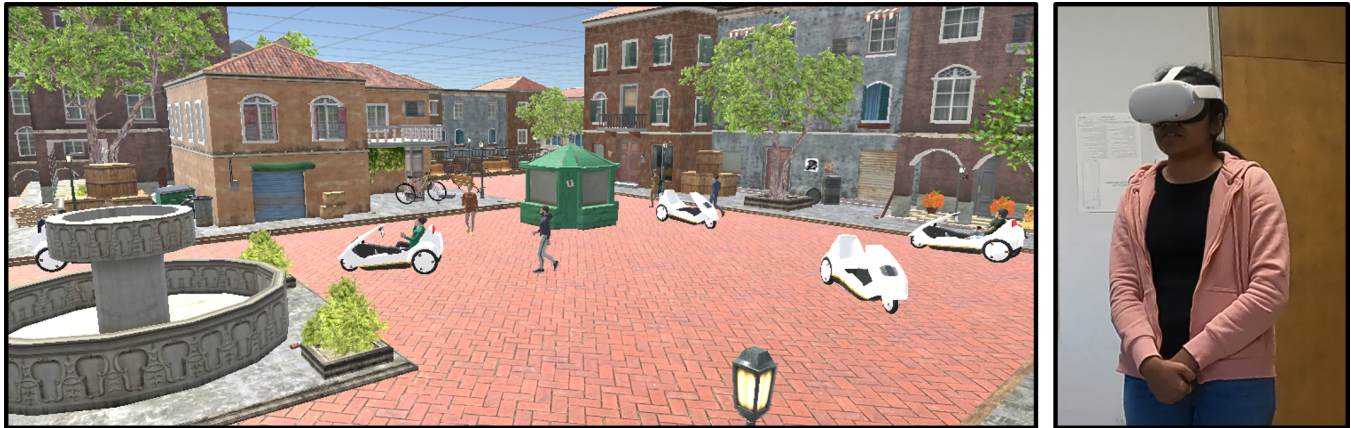


Figure 1: Study Setup: (left) a virtual reality scenario demonstrates scattered pod patterns with pedestrians in the environment; (right) a participant is testing the scene while wearing a headset.

ABSTRACT

As autonomous vehicles (AVs) advance in navigating complex settings, investigating their coexistence with people and how they are perceived becomes crucial. Pods, also known as low-speed autonomous transport systems (L-SATS), are emerging in pedestrian spaces, such as museums and malls, impacting people's comfort in these environments. This research explores how incidentally copresent persons (InCoPs) perceive pods in pedestrian spaces, focusing on their comfort levels as a preliminary factor. We conducted a virtual reality study with 10 participants, examining the importance of various variables, including pod quantity, pod group formation, passenger presence, and InCoP position. Informed by our results, we discussed insights on factors improving InCoPs' comfort, highlighting the significance of having an enhanced sense of control, the space and freedom to move around, passengers' awareness, and the social behaviors of other copresent pedestrians.

CCS CONCEPTS

• Human-centered computing → User studies.

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CHI EA '24, May 11–16, 2024, Honolulu, HI, USA
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ACM ISBN 979-8-4007-0331-7/24/05
<https://doi.org/10.1145/3613905.3650776>

KEYWORDS

Pods, copresence, pedestrians, virtual reality

ACM Reference Format:

Iryna Luchak, Ashratuz Zavin Asha, and Ehud Sharlin. 2024. Podscape: Exploring the Comfort Level with Pods in Pedestrian Spaces through Immersive Simulation. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '24)*, May 11–16, 2024, Honolulu, HI, USA. ACM, New York, NY, USA, 7 pages. <https://doi.org/10.1145/3613905.3650776>

1 INTRODUCTION

As autonomous vehicles (AVs) are becoming increasingly capable of navigating complex environments, new opportunities are emerging. One of the most pressing transportation issues today is the "first and last mile phenomenon," or the distance a commuter must travel at the beginning or the end of their trip [23]. Pedestrian spaces, including indoor spaces like malls, museums, and stadiums, contribute to the last-mile problem since walking remains the primary mode for the first and last miles of a trip [19, 30]. Autonomous vehicles in shared urban spaces, known as low-speed autonomous transport systems (L-SATS) or "pods" [35], might offer a solution to the last-mile problem within pedestrian spaces. Companies like Aurigo and A&K Robotics have developed and been testing pods in pedestrian spaces¹. The absence of a human driver allows for single-person pods similar in size to a bike or a wheelchair, as the majority of trips are done solo [5, 30]. Our work focuses on these small-sized pods, whose size, quiet movement, and low speeds make them pertinent

¹Aurigo Auto-Pod: <https://aurigo.com/autopod/>;
A&K Robotics: <https://www.aandkrobotics.com>

to most pedestrian spaces, especially benefiting those unable to drive or with reduced mobility [12, 34].

Interactions with pods extend beyond vulnerable road users (e.g., pedestrians, cyclists [11]) to any person who happens to be present in the vicinity of these pods, called incidentally copresent persons (InCoPs) [1, 29]. Findings indicate that humans are uncomfortable with empty AVs [3, 20], which may be exacerbated for InCoPs when pods enter pedestrian-reserved areas. For this reason, it is important to explore InCoPs' comfort factors when sharing the space with compact pods, giving rise to podscapes. We use the term *podscape* loosely to relate to an environment shared by pods, InCoPs, and their surroundings, as perceived by humans. While related research is conducted on autonomous vehicles in shared spaces [26], a noticeable gap exists in the literature concerning pedestrian spaces where conventional vehicles are not allowed. This underscores the need to investigate and understand the dynamics between InCoPs and pods in pedestrian spaces. Comfort is a commonly used measure in pedestrian interactions with AVs [20, 26, 36], and it influences pedestrian space quality along with safety and mobility [27]. Small-sized and slow-moving AVs have been found to increase feelings of safety [6, 9]. Thus, we focus on the comfort level of participants in our work as an initial exploration, acknowledging the necessity of evaluating other commonly measured emotions (feeling of security, trust, anxiety, etc. [8, 26, 27]) in future studies.

To the best of our knowledge, our work is the first to investigate InCoPs' experiences with pods in pedestrian spaces through a user study involving 10 participants. Utilizing virtual reality (VR), we implemented 10 scenarios, examining variables such as pod quantity, pod group formation, passenger presence, and InCoP position. The study findings contribute to identifying important factors influencing InCoPs' comfort, including pod predictability and traceability, social influence of passengers and pedestrians, freedom of movement, and limited pod dominance. Our work provides valuable insights for future research to consider when designing and deploying pods within pedestrian spaces.

2 RELATED WORK

In this section, we draw on literature from the autonomous vehicle-pedestrian domain, as well as from the mobile robots domain, as AVs can be perceived as a hybrid between a conventional vehicle and a robot [1, 26]. Our discussion of the related studies provides insight into design considerations and variables for our study.

Due to the recent development of pods, limited research exists, particularly on single-seater pods [34, 35]. Prior studies have shown that human comfort drops and intimidation rises with the size of passing and approaching robots [10], but the implications for pods are unclear. Moreover, a study investigating interactions between humans and mobile personal robots found that people feel threatened and uncomfortable by approaching robots [7], highlighting the need to study pods before introducing them into new human environments. While a VR gap acceptance study found that pedestrians feel less safe in shared spaces than on roads [35], this cannot be directly applied to pedestrian zones where conventional-sized vehicles are not permitted. Moreover, previous research studying the effects of groups and types of robots found more negative emotions (e.g., anxiety, fear, threat) when there are multiple robots, and

they appear as a single group [14, 15]. Unlike conventional roads, shared spaces have the flexibility for diverse group arrangements, which can make pods appear as a single group. Platoons, a type of group formation, have been extensively studied for feasibility [13], but seldom for the comfort of InCoPs. Focus group findings suggest platooning pods in shared spaces is generally accepted [34]. The previously discussed VR gap acceptance study found that people are more likely to cross a platoon of pods in a shared space than on roads [35]. We found no studies comparing platoon formations to individual AV movements, nor varying quantity, on people's comfort, motivating us to address these gaps.

The absence of a human, which can be interpreted as a driver or a passenger, in the AV driver's seat resulted in lower comfort scores in participants making crossing decisions in a VR study [20]. Similar to this, in a study involving elderly participants, the presence of a steering wheel in an empty AV was perceived as unusual, causing discomfort [3]. In an exploratory study with videos of AV at a crossing, distracted drivers had a negative impact on pedestrians' emotional state [36]. Thus, we focus on reviewing studies on interactions with autonomous entities, excluding studies on human-driven micro-mobility vehicles in shared spaces [37] from this context. We also exclude studies on interfaces, which have been extensively studied in the AV-pedestrian interaction domain [21]. In AV and robot studies, exploration of InCoPs' perspectives and their relative positioning is limited. We adopt the InCoP term coined in the human-robot interaction domain [29] to describe anyone who happens to be in proximity to pods, and the encounter is unintentional. InCoPs may assume various positions and roles, including sitting diners, standing tourists, and mobile dog walkers. To the best of our knowledge, InCoP adoption in AV research has not been observed before. The presence of a human walking comfortably alongside a quadruped (four-legged) robot has been shown to reduce fear and elicit positive feelings toward the robot [16]. Similarly, the behavior of other pedestrians influenced participants' crossing decisions in a VR study discussed above [20].

Overall, while research exploring pods and InCoPs in pedestrian spaces is relatively limited, there is a growing interest in this area. Our VR study investigates the impact of the following factors on InCoPs' comfort levels with pods, based on the discussed literature: pod quantity, pod group formation, passenger presence, and InCoP position. We also include walking pedestrians in all of our scenarios. To the best of our knowledge, our work is the first to explore the impact of single-seater pods on InCoPs in pedestrian spaces and examine the mentioned factors in this domain.

3 USER STUDY: METHODOLOGY

We conducted a user study with 10 participants utilizing VR simulations and explored participants' comfort in sharing pedestrian space with pods. The study was approved by the ethics committee of the authors' institution.

3.1 VR Testbed

We employ virtual reality, a common tool for assessing user experiences with autonomous vehicles [2, 18, 32], to facilitate a safe, quick, and immersive evaluation. We built our VR testbed with the Unity3D game engine and deployed it on the Oculus Quest VR

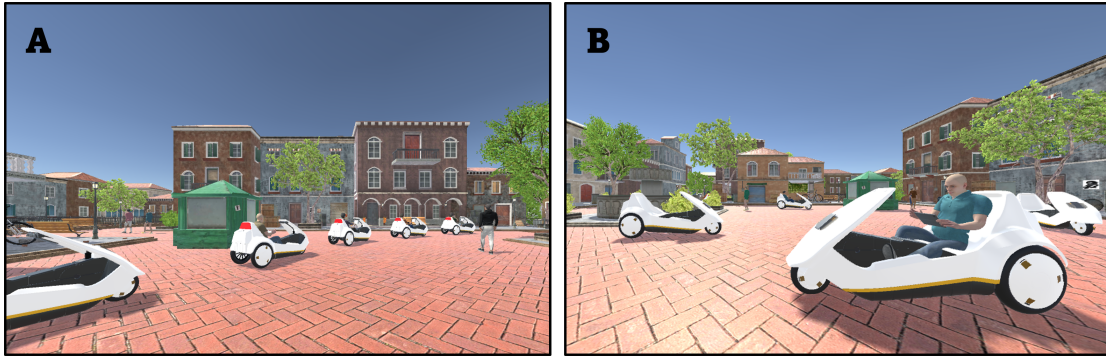


Figure 2: VR Scenario examples: (A) a scenario with platoon group formation of six pods including empty pods, (B) a scenario with scattered group formation of pods including distracted passengers.

headset. We considered the following variables, informed by literature review and brainstorming sessions: pod quantity, pod group formation, passenger presence, and InCoP position. Pod quantity includes one, three, and six pods, for baseline, average, and concentrated density comparison, respectively [14, 15]. The pod group formation can be a platoon (see Figure 2 (A)), meaning a single line arrangement [35] or scattered, indicating individual unrelated movements (see Figure 2 (B)). The passenger presence variable includes empty pods and distracted passengers as conditions [20, 36]. The InCoP position has one condition standing by a fountain and another sitting at a cafe table.

Unlike studies on traditional roads, where familiarity allows for simplified setups, we closely mimic real-life conditions to ensure the environment distinctly represents a pedestrian space and enhances participant presence and engagement [25]. We selected a suburban outdoor plaza (20 x 12 meters) as the simulation setting to establish a leisurely atmosphere and provide an open roaming space for pods and InCoPs to interact (see Figure 1 (left)). We include casual and confident walking pedestrians amongst pods [16, 20]. To maintain realism with unique characters in each scene, we modeled sixteen unique InCoPs and passengers who varied in gender, ethnicity, age, and outfits between scenarios. The pedestrian InCoPs always kept at least 1m distance from the pods with a regular walking speed of 4.3 km/h [24], which is about the distance pedestrians keep when crossing a mobile robot [33]. The pedestrians in the scene are shown as unaffected by the pods' presence and engage in collision-free navigation to reduce any bias in participants. We designed the passengers in pods to appear fully occupied with activities like reading a book (see Figure 2 (B)) to emphasize that pods are fully autonomous and do not require any human input.

The pod is a small (1.7m length x 0.8m height x 0.65m width) single-person recumbent vehicle without a steering wheel. Passengers sit in a relaxed position, emphasizing their non-driver role. The white pod color was chosen for its high visibility [20]. We settled on 5 km/h for pod speed for a modestly higher speed than pedestrians as interviews on robot acceptance revealed that pedestrians expected delivery robots to move faster than walking pedestrians [10], likely tiring of slower-than-walking speeds [7]. The pod speed is also similar to a "medium" pod speed in the VR gap acceptance study discussed earlier [35].

3.2 Study Procedure

At the beginning of each session, the study facilitator (first author) provided the background of the study. After wearing the headset (Meta Quest), we asked the participants to name three objects in the scene to help acclimate them to VR and the pedestrian space. Then, they were presented with two sets of 10 scenarios, observing one set as a standing InCoP (see Figure 1 (right)) and the other as an InCoP sitting at a cafe, for a total of 20 trials. Table 1 details each of the four independent variables and their conditions in the 20 trials. The participant took a break from the VR headset and changed the InCoP position halfway. The participants were asked to observe the moving pods and describe the comfort level of having pods in a pedestrian zone after each scenario on a Likert Scale of 1 (very uncomfortable) to 5 (very comfortable). They were also asked to reflect and explain their reasoning for the score.

Following all the trials, participants completed a semi-structured interview. We started with a set of broader questions, such as "What do you think or feel about sharing the pedestrian space with pods?" and then moved on to targeted questions, such as "How did you feel in each observation position?". We finished the interview by asking which condition they preferred for each variable (e.g., standing or sitting for the InCoP position). The study took about 45-60 minutes to complete. We randomized the order of scenarios using a Balanced Latin Square of order 10 to partially counterbalance sequencing effects within each set [28]. We also randomized the InCoP position order, with five participants starting seated and five standing, to avoid confounding effects of initial position exposure. The sessions were audio and video recorded.

3.2.1 Participants. We recruited 10 participants (4 identified as female and 6 as males; 3 undergraduate and 7 graduate students) between the ages of 20 and 31 (avg. = 25.2, SD = 3.2). We recruited participants through email and posters placed around the university. Participants received a remuneration of \$20 at the end of the study. All participants are conceptually familiar with autonomous vehicles, but only one participant has seen an autonomous vehicle on the roads in the United States. The rest of the participants have never directly seen or ridden in an autonomous vehicle. Four participants had no prior experience with virtual reality.

#	SCENARIO CONDITIONS			COMFORT SCORES																	
	Pod Quantity	Pod Group Formation	Passenger Presence	SITTING										STANDING							
				P1	P2	P3	P4	P5	P6	P8	P9	P10	P1	P2	P3	P4	P5	P6	P8	P9	P10
S1	1	N/A	passenger																		
S2	1	N/A	empty																		
S3	3	scattered	passenger																		
S4	3	platoon	passenger																		
S5	3	scattered	empty																		
S6	3	platoon	empty																		
S7	6	scattered	passenger																		
S8	6	platoon	passenger																		
S9	6	scattered	empty																		
S10	6	platoon	empty																		

1 = very uncomfortable

2 = uncomfortable

3 = neither comfortable or uncomfortable

4 = comfortable

5 = very comfortable

Table 1: Variable conditions and comfort scores heatmap for all reported 20 trials. The left side of the table displays three conditions for each of the 10 scenarios (S1 - S10). The right side of the table shows the conditions (sitting, standing) for the fourth InCoP position variable, resulting in 20 trials. Below, the corresponding heatmap of comfort scores for 10 participants (P1 - P10) is shown. P7 was excluded due to no variance in their responses (all scores were a 5).

3.3 Data Analysis

All audio recordings were fully transcribed (6 hours). Our qualitative data sources include comfort score justifications provided by participants during the study and the comments provided during the interview session. We followed thematic analysis to analyze our data [4]. We carefully read through the transcripts and let the codes develop spontaneously; some of the example codes are distance, angle of view, pod patterns, walking barrier, anxiety, collision, social norms, and road rules. Next, we iteratively analyzed the codes for similarities and grouped them into categories until themes emerged. Finally, we summarized participant preferences for each variable (e.g., platoon or scattered for the pod group formation) collected during the interview session. We excluded the quantitative analysis of comfort scores, as they yielded statistically insignificant results between conditions, likely due to a low number of participants.

4 FINDINGS

This section presents the major themes derived from our study results, focusing on 1) InCoPs' comfort with diverse pod configurations and 2) environmental factors shaping InCoPs' experiences.

4.1 InCoPs' Comfort with Diverse Pod Configurations

Participants expressed a broad range of views related to pod configurations, particularly the importance of predictable and traceable pod movements and their perception of clearance amidst the pods for free movement. They also highlighted their positioning within the pedestrian space and how it influences their field of vision and proximity to the pods.

Predictability and Traceability of Pod Movements. The perceived traceability and predictability of the pods significantly impacted participants' comfort levels. Participants found the platoon arrangement to be more organized, orderly, and easier to anticipate and track their movement, with three participants comparing it to a train. The majority of participants (9 out of 10) preferred the platoon pod group formation over scattered, with predictability as the primary cited reason. The discomfort with scattered formations arose from unpredictable trajectories and the absence of clear movement patterns, causing participants to feel anxious and apprehensive. Particularly, the 6-pod scattered scenarios resulted in negative reactions like chaos and haphazardness, highlighting the impact of increased pod numbers on discomfort.

Proximity and Field of Vision. Participants' positioning in relation to the pods and the pedestrian space influenced their comfort. Sitting further from the pods increased their angular view, enabling the participants to see more of the context, such as the entire platoon, thus enhancing their comfort level. P7 commented, "I was much further from everything in a protected space like a bubble which gave me a feeling of security". In the standing position, participants felt compelled to heighten their situational awareness for being too close to the pods, and they continuously monitored their surroundings in various directions due to the narrower field of vision. Two participants explained the feeling further, saying, "I had to look around; you know they won't hit you, but you still would check where they are going exactly." (P10). Nine out of ten participants preferred the sitting café position, and one participant felt neutral.

Freedom of Movement Amidst Pods. The comfort of participants was influenced by the impact of pods on the walkability of the

pedestrian space. Some participants perceived the platoon as a barrier to pedestrians' freedom of movement and highlighted potential disruptions to pod passengers when crossing in front of the platoon. Participants preferred waiting for pods to clear rather than disrupting the platoon's path, with one participant stating, *"I wouldn't want to disrupt their train. I would have to wait for them to get out of the way."* (P5). The majority of participants (8 out of 10) found the single-pod scenarios very comfortable. Greater numbers felt intrusive into the space and restricted the pedestrians' freedom to walk around. The 6-pod platoon was particularly seen as obstructive and a nuisance to walk around, while the scattered formation raised collision concerns. Most participants (6 out of 10) indicated that they were comfortable with a maximum of 3-4 pods in the space, regardless of the formation.

4.2 Environmental Factors Shaping InCoPs' Experiences

Participants expressed how their comfort level was influenced by other external factors in their environment: the presence of other InCoPs, particularly the walking pedestrians, the presence of passengers inside the pods, and the extent to which the pedestrian space felt preserved for its original function.

Presence of other InCoPs. The presence of walking pedestrians in the space influenced the comfort level of seven participants, while three participants reported no impact. P7 and P8 highlighted feeling safer with other pedestrians around, as being alone with self-driving cars seemed less safe. *"If I were the only one person standing there, it wouldn't feel safe within a bunch of self-driving cars."* (P8). The relaxed manner by which pedestrians moved right beside the pods without concern suggested familiarity and increased the confidence of some participants. In contrast, three participants felt anxious observing pedestrians walking casually in close proximity to the pods, fearing potential collisions.

Awareness of Pod Passengers. We found varying reactions and comfort levels in participants based on the visual appearance of the passengers (aware vs. distracted) in pods. Two participants found the empty pods shocking at first but eventually grew comfortable with them. Having passengers in pods made some participants (4 out of 10) feel more secure initially, with an added sense of control and purpose to the pods. Overall, we received mixed reactions from participants. Four participants believed passengers had minimal control over the pod's behavior and were not influenced by their presence. Three participants had the greatest confidence in empty pods, emphasizing their ability to react faster in emergencies and avoid collisions with built-in smart technologies. Three participants reported that passengers' awareness of their surroundings made them most comfortable. Overall, discomfort arose in participants when passengers were occupied with activities and not paying attention to their surroundings. P3 commented on this, saying *"[The distracted passengers] gave me an uneasy feeling. Reminded me of distracted drivers or people who don't care whether their car is about to crash into somebody."* (P8).

Pedestrian Space Integrity. Participants mentioned that the pods could compromise the integrity and disrupt the intended purpose of the pedestrian space. Three participants expressed concerns that a higher number of pods will dominate the space, stating: *"I just feel*

like there are a lot of them and instead of sharing they are taking up the space." (P5). P8 emphasized that the pedestrian space felt more natural when pedestrians outnumbered the pods. These participants also mentioned the potential disruptions posed by pods, particularly in a relaxed environment, such as adding noise, visual clutter, and more moving elements to navigate.

5 DISCUSSION

In this section, we reflect on our findings regarding InCoPs' sense of control, highlighting preferences for freedom of movement and predictable pod movements in the pedestrian space. Additionally, we explore their perception of autonomy in pods and the social influence of having pedestrians and passengers.

5.1 Sense of Control with Pods

Participants desire to feel a sense of control with pods, citing predictability and freedom of movement as the main reasons for their scenario preferences. They sought knowledge about pod behavior, including route patterns and the right of way, and preferred more distance for better visibility, instilling a sense of security. Similarly, maintaining a pedestrian-centric environment with fewer pods and ample space for InCoPs' movement enhanced the participants' feeling of control over the space. Despite being in a pedestrian-centric space, our participants were willing to yield to pods to avoid disrupting their path, similar to previous research findings where pedestrians and joggers yielded to AVs when passing in a shared space [17]. However, this contrasts with the finding that most pedestrians believed they had priority over an AV in a shared space [22]. Overall, our results inform future research with factors such as proximity, freedom of movement, and predictability influencing participants' comfort levels and positively co-existing with pods.

5.2 Perception of Autonomy in Pods

The participants' concerns about distracted passengers reveal potential trust issues in pods. This discomfort may arise from a lack of awareness about the autonomous capabilities of pods, a lingering reliance on human control, or simply being unaccustomed to the concept of a driverless vehicle. The participants' discomfort with distracted passengers prompts inquiries into the autonomy level of these vehicles and the impact of a passenger's presence, especially given artificial intelligence's potential faster reaction times than humans. These observations align with previous research where AVs with distracted passengers evoked discomfort and caution among pedestrians [36]. The visual appearance of pods has the potential to evoke safety concerns among InCoPs since passengers might be mistaken as drivers, leading to trust issues regarding the pod's autonomous mode. Similar to this, the presence of a steering wheel in an AV without a driver caused discomfort among participants in prior research [3, 20]. Notably, some participants exhibited reduced concern when the pods were empty, suggesting that as InCoPs become accustomed to pods, they may demonstrate a higher comfort level with the technology compared to human control.

5.3 Social Factors in Pedestrian Space

Social influence significantly impacted some participants' comfort and acceptance of the pods. Observing others walking close to the pods, appearing unbothered, instilled a sense of comfort among the participants. The presence of walking pedestrians reinforced the belief that pods would navigate safely without collisions, providing reassurance during emergencies. Beyond assurance, the presence of pedestrians alleviated the impact of the introduction of new technology in their pedestrian-centric environment. This aligns with previous studies, where the presence of other pedestrians influenced participants' crossing decisions [20], and a human walking beside a robot increased InCoPs' comfort levels [16]. The impact of social influence also extended to passengers, with most participants expressing a preference for pods with human occupants, regardless of their level of distraction. It created an illusion of humans exerting some control over the technology, aligning with the broader theme of overriding the automated system in emergencies [3].

5.4 Limitations and Future Work

As an initial exploration of InCoPs' comfort of being pods in the pedestrian space, there are several limitations to our study. Firstly, the interview study involved a small participant pool (N=10), potentially limiting the generalizability of our findings. However, the depth of discussions with each participant enhances the validity of our insights [31]. Given the study's scope, we had to constrain the environment and variables we tested in our implemented simulation. We limited the type of InCoP perspectives we tested, focusing on stationary InCoPs - one standing and one sitting. However, pedestrian spaces are used by moving InCoPs, such as joggers and cyclists, as well as people with mobility challenges, the elderly, and children. Future research is required to explore a broader range of pedestrian spaces and InCoP perspectives, and other factors influencing comfort, such as proxemics and social norms. Finally, the participants were North American university students, limiting cultural diversity. Future research should explore broader factors like participants' cultural backgrounds and experiences with pedestrian zones. Despite these shortcomings, our work provides preliminary reflections on having pods in pedestrian spaces and factors improving experiences for people copresent with pods.

6 CONCLUSION

Exploring the coexistence with pods in complex environments is essential, particularly considering the comfort of individuals encountering or being in proximity to them. While existing research has predominantly focused on autonomous vehicle-pedestrian interactions in shared spaces, our work delves into incidentally copresent persons' comfort with pods in pedestrian environments. We identify key factors influencing comfort, including pod configurations and freedom of mobility for InCoPs. Our findings underscore the significance of social factors, such as awareness of passengers and pedestrians displaying confidence in sharing space with pods, contributing to heightened comfort and a sense of control for InCoPs. In future extensions of our work, we aim to explore additional factors, including feelings of security, trust, and clarity, to gain a comprehensive understanding of InCoPs' experiences with pods in pedestrian spaces.

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